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Using structural equation modeling to construct calibration equations relating $PM_{2.5}$ mass concentration samplers to the federal reference method sampler

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HIGHLIGHTS

- Collocated $PM_{2.5}$ mass concentration samplers were calibrated using linked structural equation models (SEMs).
- Bias and imprecision were determined for FRM, TEOM, SFS, SASS, and IMPROVE samplers.
- SEMs were stratified to describe the temperature dependency of TEOM samplers.
- TEOM bias relative to FRM followed a well-defined sigmoidal temperature dependency.
- FRM was most precise while TEOM imprecision decreased with increasing temperature.

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ABSTRACT

The objective of this study was to remove systematic bias among fine particulate matter ($PM_{2.5}$) mass concentration measurements made by different types of samplers used in the Pittsburgh Aerosol Research and Inhalation Epidemiology Study (PARIES). PARIES is a retrospective epidemiology study that aims to provide a comprehensive analysis of the associations between air quality and human health effects in the Pittsburgh, Pennsylvania, region from 1999 to 2008. Calibration was needed in order to minimize the amount of systematic error in $PM_{2.5}$ exposure estimation as a result of including data from 97 different $PM_{2.5}$ samplers at 47 monitoring sites. Ordinary regression often has been used for calibrating air quality measurements from pairs of measurement devices; however, this is only appropriate when one of the two devices (the "independent" variable) is free from random error, which is rarely the case. A group of methods known as "errors-in-variables" (e.g., Deming regression, reduced major axis regression) has been developed to handle calibration between two devices when both are subject to random error, but these methods require information on the relative sizes of the random errors for each device, which typically cannot be obtained from the observed data. When data from more than two devices (or repeats of the same device) are available, the additional information is not used to inform the calibration. A more general approach that often has been overlooked is the use of a measurement error structural equation model (SEM) that allows the simultaneous comparison of three or more devices (or repeats). The theoretical underpinnings of all of these approaches to calibration are described, and the pros and cons of each are discussed. In particular, it is shown that both ordinary regression (when used for calibration) and Deming regression are particular examples of SEMs but with substantial deficiencies. To illustrate the use of SEMs, the 7865 daily average $PM_{2.5}$ mass concentration measurements made by seven collocated samplers at an urban monitoring site in Pittsburgh, Pennsylvania, were used. These samplers, which included three federal reference method (FRM) samplers, three speciation samplers, and a tapered element oscillating microbalance (TEOM), operated at various times during the 10-year PARIES study period. Because TEOM measurements are known to depend on temperature, the constructed SEM provided calibration equations relating the TEOM to the FRM and speciation samplers as a function of ambient temperature. It was shown that TEOM imprecision and TEOM bias (relative to the FRM) both decreased as temperature increased. It also was shown that the temperature dependency for bias was

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non-linear and followed a sigmoidal (logistic) pattern. The speciation samplers exhibited only small bias relative to the FRM samplers, although the FRM samplers were shown to be substantially more precise than both the TEOM and the speciation samplers. Comparison of the SEM results to pairwise simple linear regression results showed that the regression results can differ substantially from the correctly-derived calibration equations, especially if the less-precise device is used as the independent variable in the regression.

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1. Introduction

Several different methods are commonly used to measure ambient mass concentrations of fine particulate matter ($PM_{2.5}$), and these methods are known to exhibit systematic biases relative to one another. For example, it is well documented that $PM_{2.5}$ mass concentrations measured using a tapered element oscillating microbalance (TEOM) can be biased low relative to those measured using a federal reference method (FRM) monitor or $PM_{2.5}$ speciation sampler because of the loss of semi-volatile material, including ammonium nitrate and semi-volatile organic compounds, from the TEOM (Allen et al., 1997; Chung et al., 2001; Eatough et al., 2003; Cyrys et al., 2001; Tortajada-Genaro and Borrs, 2011). The magnitude of this bias is dependent upon the chemical composition of the $PM_{2.5}$, the TEOM operating conditions, and ambient conditions, and therefore is site- and monitor-specific. Hence, for air quality monitoring applications or scientific studies (e.g., air pollution epidemiology studies) that must incorporate or compare data from multiple types of $PM_{2.5}$ measurement devices, it is necessary to first develop calibration equations to remove systematic biases so that the data can be used interchangeably.

Ordinary linear regression analysis commonly has been employed to develop calibration equations relating $PM_{2.5}$ measurement devices. Ordinary regression analysis will provide a correct description of the relationship between two devices when it is applied to data spanning a wide range of the quantity to be measured, when one of the devices is a highly precise reference measurement (that is, one with little or no random error), and when this highly precise reference is used as the "independent" variable in the regression. In the case of calibrating $PM_{2.5}$ samplers, though, as with many applications, both devices are subject to substantial random error, even if one is termed a "reference" method. In these cases, ordinary regression produces two contradictory curves, depending on which device is used as the independent variable in the model, both of which are biased away from the correct calibration curve (Bollen, 1989). Nevertheless, numerous $PM_{2.5}$ or PM_{10} method comparison studies (e.g., Allen et al., 1997; Ayers et al., 1999; Chung et al., 2001; Charron et al., 2004; Schwab et al., 2006; Zhu et al., 2007) have been based on ordinary regression, and a number of studies (e.g., Bortnick et al., 2002; Vega et al., 2003; Kashuba and Scheff, 2008) comparing TEOM and filter-based (e.g., FRM) measurements have used the less-precise TEOM as the independent variable, exacerbating the problem. In fact, the U.S. Environmental Protection Agency (EPA) guidelines (Eberly et al., 2002) for relating FRM and continuous $PM_{2.5}$ measurements specify the use of ordinary linear regression with the presumably less-precise continuous measurement device used as the independent variable.

In the late nineteenth century, Adcock (1878), aware of the problems with using ordinary regression to calibrate two types of measurement that are both affected by random error, described a method that today is often referred to as Deming regression. Deming regression (Kummell, 1879; Deming, 1943) is a particular example of an errors-in-variables approach often applied when there are only two devices being compared, or applied in a pairwise fashion when there are more devices and/or repeats available. Several $PM_{2.5}$ method comparison studies (e.g., Hauck et al., 2004; Lee et al., 2005; Hains et al., 2007) have used this approach. In practice, however, Deming regression can be impractical to use for two reasons. Firstly, Deming regression assumes that all the error is random error and that the measurement scales are the same. Secondly, it assumes that the ratio of the imprecision standard deviations (that is, the random errors for each method) is known. Even if this ratio were known, it would be impossible to know whether the measurement scales are the same based only on the paired measurements. Pairwise application of Deming regression and other error-in-variables methods will also fail to incorporate all the available information when measurements from more than two devices are available.

Another errors-in-variable approach that has been used is reduced major axis (RMA) regression Kermack and Haldane, (1950), which is sometimes referred to as orthogonal regression. Ayers (2001) in a technical note suggested the use of RMA regression in place of ordinary regression for air quality data, and Charron et al. (2004) and Green and Fuller (2006) applied RMA regression to compare PM samplers. However, when the ratio of the device imprecision standard deviations is equal to one, Deming regression reduces to RMA regression. Thus, RMA regression suffers from the same defects as Deming regression.

Two issues should be further noted. Firstly, of the PM measurement comparison studies cited above, only Bortnick et al. (2002) transformed the PM measurements. Transformations are often required to help insure that the measurement errors are normally distributed and the relationship is linear, as assumed by most methods of analysis. Secondly, almost all of the papers used correlation coefficients as a measure of agreement, regardless of whether they used ordinary regression or not. In an ordinary regression with a single independent variable, the correlation coefficient is merely a linear transformation of the regression slope and vice versa. Importantly, the (Pearson product-moment) correlation coefficient is only a measure of linear association and does not measure agreement between devices. For example, there could be a perfect correlation between temperatures measured in Fahrenheit degrees and Celsius degrees yet, in general, there would be no agreement (in the sense that you could not substitute one for the other). Two devices can be perfectly correlated yet have no agreement and, conversely, can have a correlation of zero and yet be in good agreement. The latter can occur when the quantity being measured has a very narrow range. The correlation coefficient is highly dependent on the range of measurements – measurements made by two devices will tend to be highly correlated if the range of the true values is very large compared to the typical size of the random error in each measurement, regardless of the amount and type of bias.

The work presented here was performed as part of a larger effort to calibrate fine particulate matter ($PM_{2.5}$) mass concentration

measurements made by 97 samplers (including more than ten different types of samplers) at 47 monitoring sites used in the Pittsburgh Aerosol Research and Inhalation Epidemiology Study (PARIES). PARIES is a retrospective epidemiology study that aims to provide a comprehensive analysis of the associations between air quality and human health effects in the Pittsburgh, Pennsylvania, region from 1999 to 2008. In this paper, we focus on calibrating measurements from seven samplers, including three $PM_{2.5}$ FRM samplers, three $PM_{2.5}$ speciation samplers, and one TEOM sampler, that were collocated at a single site in the Lawrenceville neighborhood of the city of Pittsburgh. This site was chosen because it had the largest number of collocated samplers and included a TEOM sampler. (A second paper will describe how this analysis was expanded to all 47 monitoring sites). As a result of investigating various potential approaches, a structural equation model (SEM), specifically consisting of a set of linked sub-models, is proposed to appropriately describe the measurement errors associated with these various samplers. Structural equation modeling has a long history but has been primarily used in psychology, genetics, and economics to accommodate complicated measurement error problems. As will be illustrated, both errors-in-variables regression and ordinary regression are specific, relatively simple, examples of SEMs. We intend to show that SEMs are ideally suited to the problem of $PM_{2.5}$ mass concentration sampler calibration and similar calibration problems. We first describe the classic measurement error model and the basic SEM methodology and then describe ordinary regression as a SEM. Next, we discuss the problems inherent in the use of ordinary regression for calibration and the general impracticality of using errors-in-variables regression in most cases. We develop a comprehensive SEM that describes the relationships among the sampling device measurements and in particular handles TEOM calibration appropriately as a nonlinear function of temperature.

2. General approach to calibration

2.1. Classic measurement error model

A measurement error model describes how the quantity that we attempt to measure is distorted by the measurement process (Jaech, 1985). The path diagram for a one-factor measurement error model for three devices is illustrated in Fig. 1a. The sources of distortion involve both systematic error (bias, lack of accuracy, lack of trueness) and random error (imprecision, lack of repeatability, lack of reliability). Formally, each measurement Y_{ij} from device i for item j is described as:

$$Y_{ij} = \alpha_i + \beta_i \mu_j + \varepsilon_{ij} \quad (1)$$

where α_i and β_i jointly describe the bias for device i, μ_j is the true value for item j that we attempt to measure, and ε_{ij} is the random error which is usually assumed to be normally distributed with mean zero and standard deviation σ_i. The true values μ have standard deviation σ and can be treated as either fixed or random values depending on the application. This model would be an ordinary regression model except that each μ_j is in practice forever unknown, making the application of ordinary regression impossible. The slope bias parameter β characterizes the measurement scale – the size of the measurement unit – for device i relative to the true scale. The intercept bias parameter α is an arbitrary but necessary reference point and is difficult, in general, to interpret in isolation from the scale bias β. When, however, $\beta_i/\beta_{i'} = 1$, the intercept represents a constant bias between devices i and i' equal to $\alpha_i - \alpha_{i'}$ – that is, a bias that is independent of μ. When $\beta_i/\beta_{i'} \neq 1$, the bias depends on μ and the measurement units differ.

The true values represent the assumption at the heart of any calibration/measurement error analysis. Each instrument must attempt to measure the same quantity at the same time and same location. If the instruments are attempting to measure different quantities there can be no basis for calibration. For air quality studies, measurements are made on small parcels of air. Ideally, the

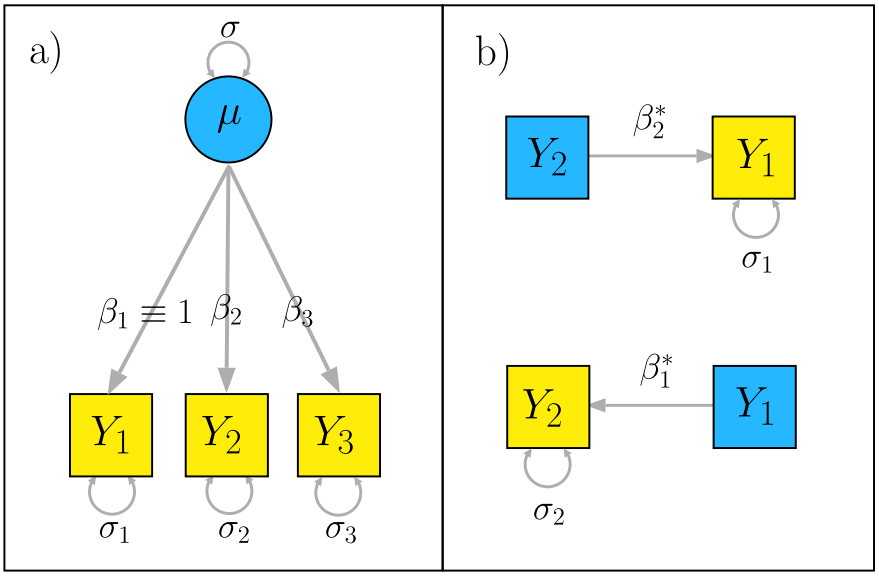


Fig. 1. Path diagrams comparing a) measurement error model for measurements Y_1, Y_2, and Y_3 from three devices 1, 2, and 3, and b) regression models for predicting measurements from device 1 (Y_1) from measurements from device 2 (Y_2), and vice versa. A circle indicates a latent variable and a square indicates a measured (observed) variable. Responses are denoted by the color yellow (light grey) while independent variables are indicated by the color blue (dark grey). The measurement error model treats the measurements as responses while the regression models must treat one of the measurements as an independent variable that has no random error. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

same parcel of air would be analyzed by all of the instruments at the same exact point in time. However, attempting to analyze the same air parcel at the same time could lead to interactions between the instruments and changes to the air parcel, so in practice slightly different air parcels are sampled with the goal of having these air parcels be as similar as possible. This usually means taking air parcels that are physically very close together and making measurements at the same instant of time. To the extent that the air parcels have different compositions, the resulting error will be misidentified as measurement error. Because the collocated samplers typically are located within a few meters of each other, the error due to misidentification is assumed to be negligible. One other point should be emphasized. Although μ is a parameter in the model, there are no unique estimates for the true values, and all determinations of bias are relative. Only highly precise reference samples, which are accurate based on theoretical understanding, can provide information on absolute accuracy.

For three samplers $i = 1$, 2, and 3 (simplifying by ignoring j), measurements Y_1, Y_2, and Y_3 are described as:

$$\begin{aligned} Y_1 &= \alpha_1 + \beta_1\mu + \varepsilon_1 \\ Y_2 &= \alpha_2 + \beta_2\mu + \varepsilon_2 \\ Y_3 &= \alpha_3 + \beta_3\mu + \varepsilon_3 \end{aligned} \tag{2}$$

with ε_1 having standard deviation σ_1, ε_2 having standard deviation σ_2, and ε_3 having standard deviation σ_3. A scale must be provided for the unobserved μ. This can be accomplished either by fixing σ to 1 and the mean of μ, $\overline{\mu}$, to zero or by arbitrarily selecting the ith device and setting β_i to one and α_i to zero. Arbitrarily setting the scale for μ to equal the scale of one of the devices is typically more convenient. The fact that μ is unknown is of little concern as the real interest is the relationship between any two devices i and i'. This can be derived easily because the two equations both contain μ, yielding the unique relationship between Y_i and $Y_{i'}$:

$$Y_i = \left[\alpha_i - \frac{\beta_i}{\beta_{i'}}\alpha_{i'}\right] + \frac{\beta_i}{\beta_{i'}}Y_{i'} + \left[\varepsilon_i - \frac{\beta_i}{\beta_{i'}}\varepsilon_{i'}\right] \tag{3}$$

or equivalently, rewritten for $Y_{i'}$ in terms of Y_i:

$$Y_{i'} = \left[\alpha_{i'} - \frac{\beta_{i'}}{\beta_i}\alpha_i\right] + \frac{\beta_{i'}}{\beta_i}Y_i + \left[\varepsilon_{i'} - \frac{\beta_{i'}}{\beta_i}\varepsilon_i\right] \tag{4}$$

with

$$\varepsilon_i - \frac{\beta_i}{\beta_{i'}}\varepsilon_{i'} \sim N\left(0, \sigma_i^2 + \frac{\beta_i^2}{\beta_{i'}^2}\sigma_{i'}^2\right) \tag{5}$$

and

$$\varepsilon_{i'} - \frac{\beta_{i'}}{\beta_i}\varepsilon_i \sim N\left(0, \sigma_{i'}^2 + \frac{\beta_{i'}^2}{\beta_i^2}\sigma_i^2\right). \tag{6}$$

Thus, in principle, because the random errors vanish via expectation, it is possible to remove bias between devices – that is, to calibrate one device in terms of the other. When $\beta_i/\beta_{i'} \neq 1$, the devices have different measurement scales, and this is referred to as a scale bias or equivalently as a non-constant bias, because the bias is a function of the true quantity μ. When $\beta_i/\beta_{i'} = 1$ and $\alpha_i - \alpha_{i'} \neq 0$, there is a constant bias. As noted by Dunn (2004), the random error is gauged by comparing the imprecision standard deviations adjusted for differences in scale, namely: σ_i/β_i and $\sigma_{i'}/\beta_{i'}$. Bias can be removed by calibration, but differences in imprecision can only be diminished by averaging two or more independent repeats.

Although the values for β_i and $\beta_{i'}$ are dependent on the choice of a referent device, the ratio $\beta_i/\beta_{i'}$ is invariant to this choice. Similarly, the difference $\alpha_i - (\beta_i/\beta_{i'})\alpha_{i'}$ is invariant, but note that this difference can only be easily interpreted when $\beta_i/\beta_{i'} = 1$. Given that either $\sigma \equiv 1$ and $\overline{\mu} \equiv 0$, or for one arbitrarily selected device i, $\beta_i \equiv 1$ and $\alpha_i \equiv 0$, then for $m \geq 1$ devices, there are $3m$ parameters that need to be estimated. From the sample data, there would be $m(m+3)/2$ sample variances, covariances, and means. (Under the assumption of normally distributed true values and errors, these statistics would represent the "sufficient" statistics.)

This model can be easily estimated using either the Method of Moments or the Method of Maximum Likelihood, provided there are at least $m \geq 3$ devices. Typically, maximum likelihood estimates would be preferred, especially when there are missing values, using full information maximum likelihood. In the Method of Moments estimation, the sample moments are equated to the theoretical moments (which are derived from the model). With three devices there would be nine equations and nine unknowns. Solving for the nine unknowns provides the parameter estimates.

2.2. Structural equation model

The measurement error model for three samplers is an example of a simple structural equation model (SEM) which has a path diagram shown in Fig. 1a (Bollen, 1989; Kline, 1998; Marcoulides and Moustaki, 2002). When there are fewer than three devices, the measurement error model has no unique solutions, and the SEM is said to be non-identifiable. It can easily be seen that when there are only two samplers, if one assumes that the ratio of the device random error standard deviations σ_1/σ_2 is known, then this effectively eliminates one parameter and a unique solution can be found. Some errors-in-variables models for the comparison of exactly two devices depend on this constraint to achieve a unique solution; however, a unique solution also can be achieved by making other assumptions. For example, one could assume that $\beta_1 = \beta_2$, or more generally that $\beta_1 = k\beta_2$ with $k \geq 0$. Arbitrary assumptions should be avoided and all parameters estimated from the available data from an appropriately designed study. If sufficient information is not available within the measurement error study, then the random error standard deviations could be estimated via an external study in which repeat measurements are made with both devices under identical conditions. To the extent that the externally estimated random error parameters differ from what they should be for the calibration study, this will distort estimation of the bias parameters.

2.3. Regression as a SEM

Ordinary regression is one of the simplest examples of a structural equation model, consisting of only a single equation, but it requires declaring one of the measurements a dependent (response) variable and the other an independent (explanatory) variable. If Y_1 is (arbitrarily) chosen as the independent variable and Y_2 is chosen as the dependent variable, then the path diagram is shown in the bottom of Fig. 1b and:

$$Y_2 = \alpha_2^* + \beta_2^* Y_1 + \varepsilon_2^* \tag{7}$$

$$\varepsilon_2^* \sim N\left(0, \sigma_2^{*2}\right) \tag{8}$$

where the asterisks emphasize that these parameters are not necessarily the same (and in fact, generally are different) than in the classical measurement error model described previously. Compare Equation (7) with the calibration equation (derived from Equation (3) with $i'=1$, $i=2$, $\alpha_1 = 0$, and $\beta_1 = 1$):

$$Y_2 = \alpha_2 + \beta_2 Y_1 + [\varepsilon_2 - \beta_2 \varepsilon_1]. \quad (9)$$

It is clear that the error terms differ and the calibration equation has two sources of error rather than the one source of error for regression. If instead, Y_2 is chosen as the independent variable and Y_1 is chosen as the dependent variable, then the path diagram is shown in the top of Fig. 1b and results in an entirely different equation:

$$Y_1 = \alpha_1^* + \beta_1^* Y_2 + \varepsilon_1^* \quad (10)$$

$$\varepsilon_1^* \sim N\left(0, \sigma_1^{*2}\right). \quad (11)$$

Compare Equation (10) with the calibration equation written for Y_1 as a function of Y_2:

$$Y_1 = -\frac{\alpha_2}{\beta_2} + \frac{1}{\beta_2} Y_2 + \left[\varepsilon_1 - \frac{\varepsilon_2}{\beta_2}\right]. \quad (12)$$

Note that, unlike the two distinct regression equations, Equations (9) and 12 are re-expressions of the same calibration equation. Note also that the slopes for the two ways to express the calibration are reciprocals of each other – this is not generally the case for the two different regression equations.

2.4. Why and how ordinary regression fails for calibration

A fundamental assumption of ordinary regression is that the independent variable is free from random measurement error. The regression path diagrams shown in Fig. 1b emphasize that only the response variable is subject to random error – the independent variable has no random error. When this is not the case, as mentioned previously, it is well-known that the regression slope is biased toward zero, such that if β_1 is constrained to equal 1, then:

$$\beta_2^* < \frac{\beta_2}{\beta_1} = \frac{\beta_2}{1} = \beta_2 \quad (13)$$

and

$$\beta_1^* < \frac{\beta_1}{\beta_2} = \frac{1}{\beta_2}. \quad (14)$$

For example, if there were no bias such that $\alpha_1 = \alpha_2 = 0$ and $\beta_1 = \beta_2 = 1$ so that $Y_1 = Y_2$, then in general $\beta_1^* \neq \beta_2^* \neq 1$ and $\alpha_1^* \neq \alpha_2^* \neq 0$, and regression will wrongly indicate a bias. This bias increases as the magnitude of the correlation between Y_1 and Y_2 decreases (Bollen, 1989).

In general, the calibration line will fall between the two regression lines. If $\sigma_1/\beta_1 = \sigma_2/\beta_2$, then the calibration curve will fall equidistantly from the two regression lines. For a fixed $\sigma_1 \neq 0$, as σ_2 approaches zero, the calibration line will approach the regression line for Y_1 as a function of Y_2. Similarly, for a fixed $\sigma_2 \neq 0$, as σ_1 approaches zero, the calibration line will approach the regression line for Y_2 as a function of Y_1.

Therefore, in practice, ordinary regression is only suitable for calibration when one of the two measurement devices being compared can be considered to have essentially no random measurement error compared with the other device, and this first device is used as the independent variable in the regression.

3. Methods

3.1. $PM_{2.5}$ mass concentration measurements

Structural equation modeling was applied to calibrate collocated $PM_{2.5}$ mass concentration measurements that were made between 1999 and 2008 at a monitoring site in Pittsburgh, Pennsylvania. The site (40.466 N, 79.961 W), which is maintained by the Allegheny County Health Department (ACHD), is located in Pittsburgh's Lawrenceville neighborhood, an urban area about two miles northeast of downtown. Seven $PM_{2.5}$ samplers, run by three different monitoring organizations, operated at the Lawrenceville site during some or all of the 1999–2008 period. These samplers included three $PM_{2.5}$ FRM samplers, three $PM_{2.5}$ speciation samplers, and one TEOM.

The primary FRM sampler (FRM 1) at the Lawrenceville site was a Rupprecht and Patashnick (R&P) Partisol-Plus Model 2025 Sequential Air Sampler operated by the ACHD. This sampler measured $PM_{2.5}$ mass concentrations on an approximately daily basis during February 1999–December 2008. The ACHD operated a second R&P FRM sampler (FRM 2) at the site on a 1-in-6-day frequency between January 2005 and December 2008 to provide collocated data. In addition, a third R&P FRM sampler (FRM 3) was operated approximately every sixth day between February 1999 and August 2001 by the Upper Ohio River Valley Project (UORVP), which stationed some of its air monitoring equipment at the Lawrenceville site.

The three $PM_{2.5}$ speciation samplers at the Lawrenceville site included a Met One Spiral Aerosol Speciation Sampler (SASS) operated by the ACHD, a Desert Research Institute (DRI) Sequential Filter Sampler (SFS) operated by the UORVP, and an Interagency Monitoring of Protected Visual Environments (IMPROVE) sampler. The SASS sampler measured $PM_{2.5}$ mass concentrations on an approximately 1-in-3-day schedule between June 2001 and December 2008, and the IMPROVE sampler measured $PM_{2.5}$ mass concentrations on a 1-in-3-day schedule between April 2004 and June 2005. The SFS sampler generally operated every sixth day between February 1999 and August 2001, except that it operated on a daily basis during several sampling intensives within this period and during additional intensives in January 2002 and October 2002–February 2003.

In all of the FRM and speciation samplers, integrated $PM_{2.5}$ samples were collected on Teflon filters at ambient temperature and pressure. Each filter was conditioned in the laboratory before and after exposure under controlled conditions of temperature and humidity, and the mass of sampled $PM_{2.5}$ was determined gravimetrically. The average $PM_{2.5}$ concentration during the sampling period was then determined based upon the mass of sample collected and the volume of air sampled. Although the FRM and speciation samplers employ a common measurement principle, slight differences in performance are possible because of differences in sampler design. For example, the R&P FRM samplers were equipped with WINS impactors to separate $PM_{2.5}$ from larger particles and sampled ambient air at a flow rate of 16.7 L/min. The SASS, SFS, and IMPROVE samplers were equipped with cyclones for particle size separation and sampled at flow rates of 6.7, 20, and 22.8 L/min, respectively. The FRM and speciation samplers also differed in their flow control schemes. These variations in sampler design result in different filter face velocities, size cut points, and particle collection efficiencies, which can lead to discrepancies in measured $PM_{2.5}$ mass concentrations (Hains et al., 2007; Chow et al., 2006; U.S. Environmental Protection Agency, 2001).

The R&P TEOM sampler at the Lawrenceville site was operated by ACHD and measured $PM_{2.5}$ mass concentrations on a continuous basis during May 2000–December 2008. In the TEOM, 16.7 L/min of

ambient air is drawn through a sharp-cut cyclone to selectively sample $PM_{2.5}$, and a 3 L/min stream of this sampled air is then drawn through a Teflon-coated glass fiber filter that sits atop an oscillating hollow tube. The frequency of oscillation changes as a function of the mass of $PM_{2.5}$ that has accumulated on the filter, enabling a near real-time determination of the concentration of $PM_{2.5}$ in the sampled air. Because the TEOM is a continuous monitor, the filter cannot be conditioned in the laboratory to control moisture, as is done in the manual sampling methods described above. Rather, the TEOM filter and sample stream were heated to 50 °C to remove moisture from the sampled air and prevent condensation on the filter. At the Lawrenceville site, the TEOM was configured to automatically adjust $PM_{2.5}$ mass concentrations, using a linear transformation with slope = 1.03 and intercept = 3.0, in order to partially account for moisture equilibration differences relative to the manual sampling methods (Thermo Scientific, 2007). However, even with this correction, additional bias can arise from the loss of semi-volatile particulate matter (e.g., ammonium nitrate, semi-volatile organic compounds) at the elevated TEOM operating temperature, causing an underestimation of total $PM_{2.5}$ mass relative to the manual samplers (Allen et al., 1997; Ayers et al., 1999). This bias is explored in depth in this paper.

Data from the ACHD FRM samplers, SASS sampler, and TEOM sampler were obtained from the U.S. Environmental Protection Agency (EPA) Air Quality System (Technology Transfer Network Air Quality System, 2010). Data from the UORVP FRM sampler and SFS sampler were obtained from the UORVP project team, and data from the IMPROVE sampler were obtained from the Visibility Information Exchange Web System (Views Data Wizard, 2010). The quality assurance and quality control (QA/QC) procedures applied to each dataset were reviewed, and a common set of data validation flags were applied to all of the data to provide for consistency. Invalid data identified in the native datasets were excluded from the analysis, as were 27 outliers identified in the native datasets and one outlier identified upon subsequent QA/QC review. In general, the FRM and speciation samplers measured 24-h average $PM_{2.5}$ mass concentrations from midnight to midnight. However, the SFS sampler measured 6-h average concentrations during the sampling intensives identified above. These 6-h data, as well as the 1-h concentrations reported by the TEOM, were averaged to compute 24-h, midnight-to-midnight averages for comparison with the other data. These 24-h averages were considered to be valid only if they included at least 19 h of valid measurement time. Table 1 presents descriptive statistics for the 24-hr average $PM_{2.5}$ mass concentrations determined by each monitor at the Lawrenceville site.

In addition to the $PM_{2.5}$ mass concentration data, daily mean temperature data from the Pittsburgh International Airport weather station (40.501 N, 80.231 W – 23.16 km from the Lawrenceville site) were used to explore the effect of ambient temperature on the calibration results. These data were obtained from the National Climatic Data Center (National Climatic Data Center, 2011).

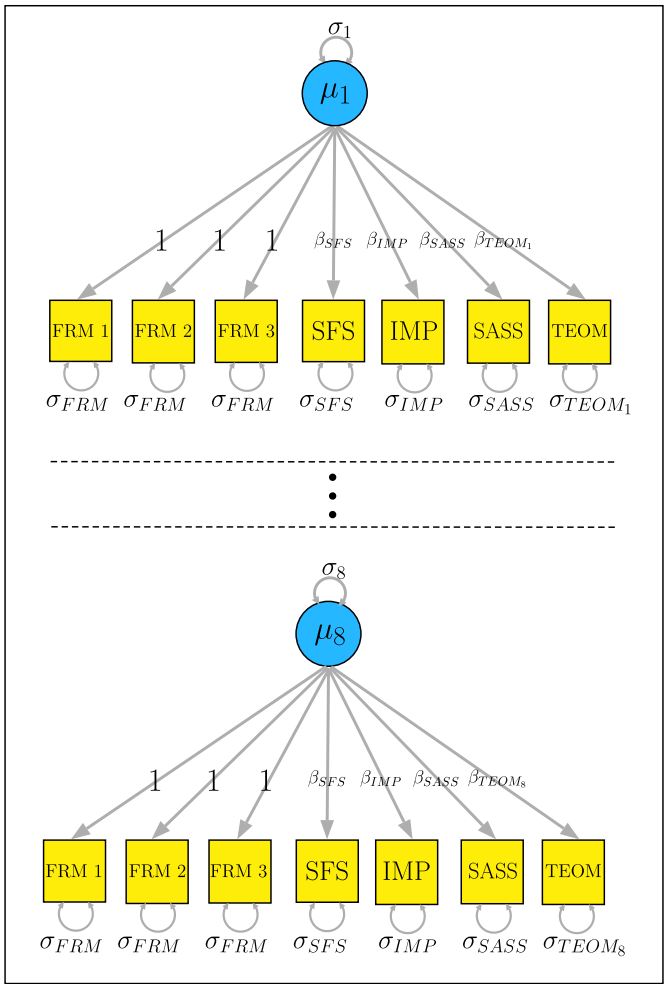


Fig. 2. Path diagram for structural equation model for measurement error relating all seven samplers and stratified by temperature. This model is composed of eight submodels with some parameters constrained to be equal across the temperature strata.

Table 1
Summary of 24-h average $PM_{2.5}$ mass concentrations determined by the collocated samplers at the Lawrenceville site.

Sampler type	Monitoring program	Abbr.	Sampling period	*N*	μg/m^3				
					Mean	SD	Min	Median	Max
R&P FRM	ACHD	FRM 1	2/1999–12/2008	3125	15.2	8.5	2.0	13.4	63.9
R&P FRM	ACHD	FRM 2	1/2005–12/2008	228	14.5	8.1	2.4	13.3	58.2
R&P FRM	UORVP	FRM 3	2/1999–8/2001	129	17.1	10.8	3.7	14.8	65.2
DRI SFS[a]	UORVP	SFS	2/1999–2/2003	439	14.7	8.5	2.6	13.2	74.1
IMPROVE	IMPROVE	IMPROVE	4/2004–6/2005	122	14.8	8.5	3.0	13.0	48.0
Met One SASS	ACHD	SASS	6/2001–12/2008	766	16.9	8.9	3.1	14.7	62.1
R&P TEOM 50 °C[b]	ACHD	TEOM	5/2000–12/2008	3056	13.8	8.8	0.0	11.8	57.5

R&P = Rupprecht & Patashnick, FRM = Federal Reference Method, ACHD = Allegheny County Health Department, UORVP = Upper Ohio River Valley Project, SASS = Spiral Aerosol Speciation Sampler, DRI = Desert Research Institute, SFS = Sequential Filter Sampler, IMPROVE = Interagency Monitoring of Protected Visual Environments, TEOM = Tapered Element Oscillating Microbalance.

[a] Certain 24-h average concentrations were derived by averaging 6-h data.

[b] 24-h average concentrations were derived by averaging 1-h data.

3.2. Statistical modeling

The R language and environment for statistical computations (version 2.13.1) was used for all computations and graphics (R Core Team, 2013). The OpenMx R package (version 1.1.2-1818) was used to describe the SEM and estimate the model parameters using full information maximum likelihood (FIML) (Boker et al., 2011, 2012). The OpenMx package also provided the ability to compute likelihood-based confidence intervals for any and all desired functions of the model parameters.

Preliminary analyses showed that the $PM_{2.5}$ mass concentration measurements tended to have distributions that were positively skewed. Further investigation showed that the log transformation tended to produce negatively-skewed distributions while the square root transformation produced more normally distributed values. All SEM computations were therefore applied to square root transformed values, and the results, where needed, were transformed back to the original scale.

Because the bias between $PM_{2.5}$ measurement devices has been shown to vary with temperature and/or season, the SEM developed for the Lawrenceville samplers allowed for an effect of temperature on the calibration results. Fig. 2 shows the path diagram for a temperature-stratified SEM that includes all seven samplers at the Lawrenceville site. There are eight sub-models: one for each temperature stratum (only the lowest and highest temperature strata are explicitly shown). In each sub-model, the FRMs are (arbitrarily) used as the reference. (As explained previously, the relevant parameter functions are invariant to this selection.) Each sub-model could be estimated separately. However, extensive analyses using these types of models indicated, as would be expected, that only the TEOM bias and imprecision parameters depended on temperature. Because these parameters for the other samplers were not temperature-dependent, the model was simplified by equating their parameters across temperature strata. Only the TEOM bias and imprecision parameters were allowed to vary with temperature in this model, and thus the overall model is composed of eight linked sub-models. Not all seven samplers operated simultaneously, and in fact, there were never more than a total of five collocated samplers operating on any given day. Therefore, the dataset of collocated measurements includes numerous missing values. However, there was enough overlap in the dataset to allow for estimation of the agreement of every sampler with all six other samplers. Because of the missing values, the SEM parameters cannot be estimated by simply using the information contained in the covariance matrix and means vectors. Instead, the solution was computed using FIML estimation using the raw data. In this method, the likelihood function was computed for each daily set of collocated measurements (including those with missing values). They provide less independent information.

4. Results and discussion

4.1. Description of the pattern of measurement collocation

Table 2 shows the number of measurements for each type of sampler and the number of collocated measurements for each possible pair of samplers. For example, there were 3125 FRM 1 measurements, with 220 collocated pairs with FRM 2 and 87 pairs with FRM 3. FRM 2 and FRM 3 measurements were never collocated (so the number of pairs was zero). For the speciation samplers, SFS and SASS had 96 pairs as did IMPROVE and SASS. SFS was never collocated with IMPROVE. For TEOM, there were a total of 3056 measurements. Of these, 2907 were collocated with FRM 1, 223 with FRM 2, and 66 with FRM 3. The TEOM measurements were split into eight temperature strata having means that ranged from −5.8 °C to 24.4 °C. The number of measurements in each stratum ranged from 359 (for −5.8 °C) to 407 (18.1 °C). Of the 372 TEOM measurements in the 14.4 °C stratum, 347 were collocated with FRM and 97 were collocated with SASS.

4.2. SEM parameter estimates

The first SEM included separate bias and imprecision estimates for the FRM samplers (1, 2, and 3). These results are shown in Table 3. For FRM 1 and 2, there was essentially no scale bias and no evidence of any constant bias, and the bias estimates are not significantly different ($\alpha = 0.05$) from a no-bias model. FRM 3

Table 3
Comparison of measurement error among FRM 1, FRM 2, and FRM 3 on the square root transformed scale. Point estimates and corresponding 95% confidence interval bounds are shown. For scale bias and imprecision SD ratios, estimates shown in bold are statistically significantly different from one.

FRM	Parameter	Lower	Estimate	Upper
Scale-adjusted intercept differences				
1	α_1 (fixed)		0.00	
2 to 1	$\alpha_2 - \frac{\beta_2}{\beta_1}\alpha_1$	−0.04	0.01	0.06
3 to 1	$\alpha_3 - \frac{\beta_3}{\beta_1}\alpha_1$	−0.48	−0.32	−0.15
3 to 2	$\alpha_3 - \frac{\beta_3}{\beta_2}\alpha_2$	−0.50	−0.33	−0.15
Scale bias				
1	β_{1P} (fixed)		1.00	
2 to 1	β_2/β_1	0.98	0.99	1.01
3 to 1	β_3/β_1	1.02	**1.06**	1.10
3 to 2	β_3/β_2	1.03	**1.07**	1.11
Scale-adjusted imprecision SDs				
1	σ_1/β_1	0.08	0.10	0.11
2	σ_2/β_2	0.02	0.05	0.08
3	σ_3/β_3	0.15	0.18	0.23
Imprecision SD ratios				
2 to 1	σ_2/σ_1	0.20	**0.56**	0.91
3 to 1	σ_3/σ_1	1.47	**1.91**	2.48
3 to 2	σ_3/σ_2	2.24	**3.40**	18.99

Table 2
Sample sizes for each sampler and pairs of sampler for $PM_{2.5}$ mass concentration measurements used in the structural equation model for measurement error at the Lawrenceville site. For the TEOM, sample sizes for each of the eight temperature strata are also shown.

		FRM			Speciation			TEOM (°C)								
		1	2	3	SFS	IMP	SASS	All	−5.8	0.7	5.3	10.0	14.4	18.1	21.1	24.4
FRM	1	3125	220	87	338	114	737	2907	343	379	358	352	347	394	369	365
	2		228	0	0	17	192	223	20	29	27	32	33	26	27	29
	3			129	123	0	6	66	7	9	5	5	8	11	14	7
Spec.	SFS				439	0	96	292	65	55	42	18	19	24	38	31
	IMP					122	96	120	12	9	18	13	16	22	16	14
	SASS						766	749	85	86	94	85	97	88	104	110
TEOM								3056	359	400	380	369	372	407	387	382

consisted of only a small number of measurements (87) collocated with FRM 1, and a small but statistically significant bias was detected relative to FRM 1 and 2. The imprecision (on the square root scale) was similar among the three FRM samplers in comparison to the non-FRM samplers as will be shown. There was enough information, however, to detect small absolute differences in imprecision (i.e., the ratios were statistically significantly different from 1). FRM 2 had an imprecision standard deviation (after correction for the tiny scale bias) that was at least 9% lower than FRM 1. The imprecision for FRM 3 (after correction for the somewhat larger scale bias) was at least 47% higher than that for FRM 1. These small biases and small differences in absolute imprecision were ignored for the final analysis so that the three FRMs were used interchangeably.

The SEM parameter estimates (relative to the FRM) for the speciation samplers and TEOM are shown in Table 4. The included corresponding 95% confidence intervals indicate that the parameter estimates are reliable (the interval widths are reasonably narrow). The results in Table 4 clearly show that the precision of the FRM sampler is superior to that of the other samplers. Among the speciation samplers, the IMPROVE and SASS samplers had comparable imprecision standard deviations (1.92 and 1.96, respectively, times that of the FRM), while the SFS sampler was considerably less precise (imprecision was 4.90 times that of the FRM). The TEOM measurements made near −5.8 °C (the lowest temperature stratum) had the highest imprecision of all samplers (about 7 times that of the FRM). The imprecision of the TEOM improves with increasing temperature, but even at the highest temperatures (about 24.4 °C) the TEOM is much less precise than the FRM (imprecision ratio = 2.94). For every comparison presented in Table 4, the imprecision ratios were statistically significant as shown by the corresponding confidence intervals.

For systematic bias compared to the FRM, Table 4 shows that the IMPROVE and SASS samplers may have no scale bias, as the confidence intervals include 1, and the point estimates are near 1. Assuming that there is no scale bias ($\beta = 1$), the intercepts for these two samplers are near zero although the interval for the SASS sampler is narrow enough to exclude zero. The SFS scale bias relative to the FRM is estimated as 0.89, with a 95% confidence interval of 0.85–0.93.

The results in Table 4 can be used to write the calibration equations to remove the relative bias between the SFS and FRM samplers. To emphasize that the parameters are relative to the FRM, let

$$\beta_{i/FRM} = \frac{\beta_i}{\beta_{FRM}} = \beta_i \tag{15}$$

and

$$\alpha_{i/FRM} = \alpha_i - \frac{\beta_i}{\beta_{FRM}}\alpha_{FRM} = \alpha_i. \tag{16}$$

In this model, the FRM was arbitrarily used as the reference (α_{FRM} fixed to 0 and β_{FRM} fixed to 1) and measurements were stratified into eight temperature strata (having roughly equal sample sizes) with the stratum mean temperatures indicated. Because the FRM was used as the reference, the relationship between each sampler and the FRM is simply:

$$\sqrt{Sampler_i} = \alpha_{i/FRM} + \beta_{i/FRM}\sqrt{FRM} + \varepsilon_i - \beta_{i/FRM}\varepsilon_{FRM} \tag{17}$$

$$\sqrt{FRM} = -\frac{\alpha_{i/FRM}}{\beta_{i/FRM}} + \frac{1}{\beta_{i/FRM}}\sqrt{Sampler_i} + \varepsilon_{FRM} - \frac{\varepsilon_i}{\beta_{i/FRM}} \tag{18}$$

where ε_i is the normally distributed random error with mean zero and standard deviation σ_i and ε_{FRM} is normally distributed with mean zero and standard deviation σ_{FRM}. In this model, σ_{FRM} is measured on the same scale as μ and σ_i is measured on the scale $\beta_i\mu$. When the sampler measurement scales differ from the FRM (i.e., $\beta_i/\beta_{FRM} \neq 1$), then to compare the imprecision of the two devices, it is necessary to compare σ_i/β_i to σ_{FRM}. The calibration equation linking *SFS* and *FRM* after taking expectations and solving for $i = SFS$ is then given by:

$$SFS = \left(\alpha_{SFS/FRM} + \beta_{SFS/FRM}\sqrt{FRM}\right)^2 = \left(0.25 + 0.89\sqrt{FRM}\right)^2. \tag{19}$$

Equivalently, solving the same equation for *FRM*:

$$FRM = \left(-\frac{\alpha_{SFS/FRM}}{\beta_{SFS/FRM}} + \frac{1}{\beta_{SFS/FRM}}\sqrt{SFS}\right)^2 = \left(-0.28 + 1.12\sqrt{SFS}\right)^2. \tag{20}$$

The calibration equations relating SASS and IMPROVE with FRM can be similarly derived and along with SFS are illustrated in Fig. 3. The

Table 4
Parameter estimates and confidence intervals computed using the structural equation model for $PM_{2.5}$ mass concentration measurement error at the Lawrenceville site. The measurements were square root transformed to achieve approximately Normal distribution, and the estimates are shown on the square root transformed scale.

Sampler	Temp. °C	Bias (systematic error)						Random error			Ratio of Adj. imprecision SD to FRM		
		Slope (scale)			Intercept (location)			Scale adjusted imprecision SD					
		$\frac{\beta_i}{\beta_{FRM}}$			$\alpha_i - \frac{\beta_i}{\beta_{FRM}}\alpha_{FRM}$			$\frac{\sigma_i}{\beta_i}$			$\frac{\sigma_i/\beta_i}{\sigma_{FRM}}$		
		Lower	Est.	Upper	Lower	Est.	Upper	Lower	Est.	Upper	Lower	Est.	Upper
FRM 1–3			1.00			0.00		0.10	0.10	0.11		1.00	
SFS		0.85	0.89	0.93	0.08	0.25	0.42	0.46	0.50	0.55	4.37	4.90	5.52
IMPROVE		0.98	1.02	1.06	−0.24	−0.09	0.07	0.17	0.20	0.23	1.59	1.92	2.31
SASS		0.97	0.98	1.00	0.18	0.24	0.30	0.19	0.20	0.21	1.76	1.96	2.17
TEOM	−5.8	0.63	0.70	0.77	0.19	0.45	0.72	0.63	0.72	0.83	6.04	7.00	8.21
	0.7	0.67	0.72	0.77	0.34	0.52	0.70	0.51	0.57	0.63	4.86	5.52	6.28
	5.3	0.72	0.78	0.83	0.28	0.46	0.64	0.49	0.54	0.60	4.65	5.27	6.02
	10.0	0.79	0.86	0.93	0.09	0.31	0.53	0.49	0.54	0.61	4.63	5.29	6.07
	14.4	0.89	0.94	1.00	−0.07	0.12	0.31	0.41	0.45	0.49	3.84	4.35	4.94
	18.1	0.96	1.01	1.06	−0.24	−0.05	0.15	0.38	0.42	0.46	3.60	4.06	4.58
	21.1	1.01	1.05	1.09	−0.38	−0.28	−0.09	0.34	0.38	0.41	3.25	3.66	4.12
	24.4	0.99	1.02	1.05	−0.35	−0.18	−0.01	0.28	0.30	0.33	2.61	2.94	3.31

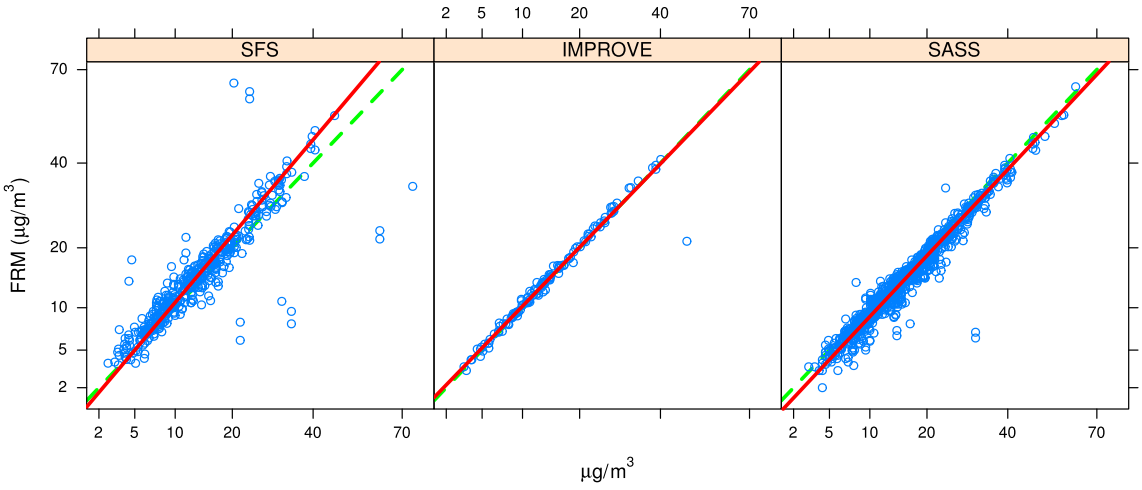


Fig. 3. Calibration curves (solid lines) for the FRM as a function of the speciation samplers SFS, IMPROVE, and SASS. The no bias model is denoted by the dashed line. IMPROVE shows essentially no bias while the SFS shows a scale bias.

scale bias for the TEOM as shown in Table 4 and Fig. 4 diminishes (β_i/β_{FRM} approaches 1) as the temperature increases, while the intercept ($\alpha_i - \beta_i/\beta_{FRM}\alpha_{FRM}$) trends from positive to zero and then to negative. Four-parameter logistic functions were fitted to the intercepts (α_T) and to the slopes (β_T) for $\sqrt{TEOM}$ as a function of $\sqrt{FRM}$ across temperature T using weighted non-linear regression. (Modeling was performed using the `SSfpl` self-starting R function with the `nls` R function for nonlinear regression.) A weighted linear regression was used to fit the $\sqrt{TEOM}$ scale-adjusted imprecision variances (σ_T/β_T) as a linear function of temperature. The regression models provided a way to pool information across the eight temperature strata and to interpolate between the observed temperature stratum means. The weights were equal to the reciprocal of confidence interval widths for each dependent variable (α_T, β_T, or σ_T/β_T). Fig. 5 shows that the relationship for both the scale bias and the intercept are non-linear and follow sigmoidal (logistic) patterns, whereas the imprecision variance decreases in a linear fashion. The corresponding parameter estimates are summarized in Table 5. The calibration equation for relating the TEOM and the FRM can be expressed as:

$$\sqrt{TEOM} = \alpha(T) + \beta(T)\sqrt{FRM} + \left[\varepsilon_{TEOM}(T) - \beta(T)\varepsilon_{FRM}(T)\right] \qquad (21)$$

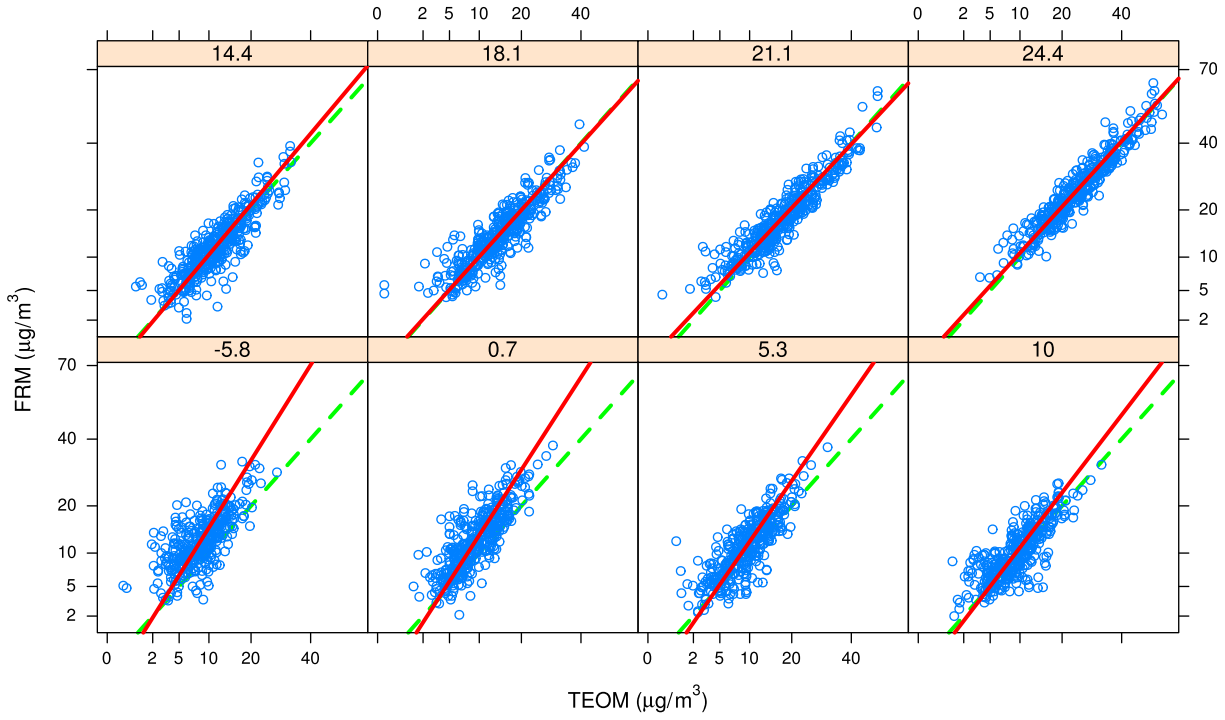


Fig. 4. Calibration curves (solid lines) for FRM as a function of TEOM stratified by temperature in °C. The mean temperature for each stratum is shown above the corresponding plot. The no bias model is denoted by the dashed line.

with expectation:

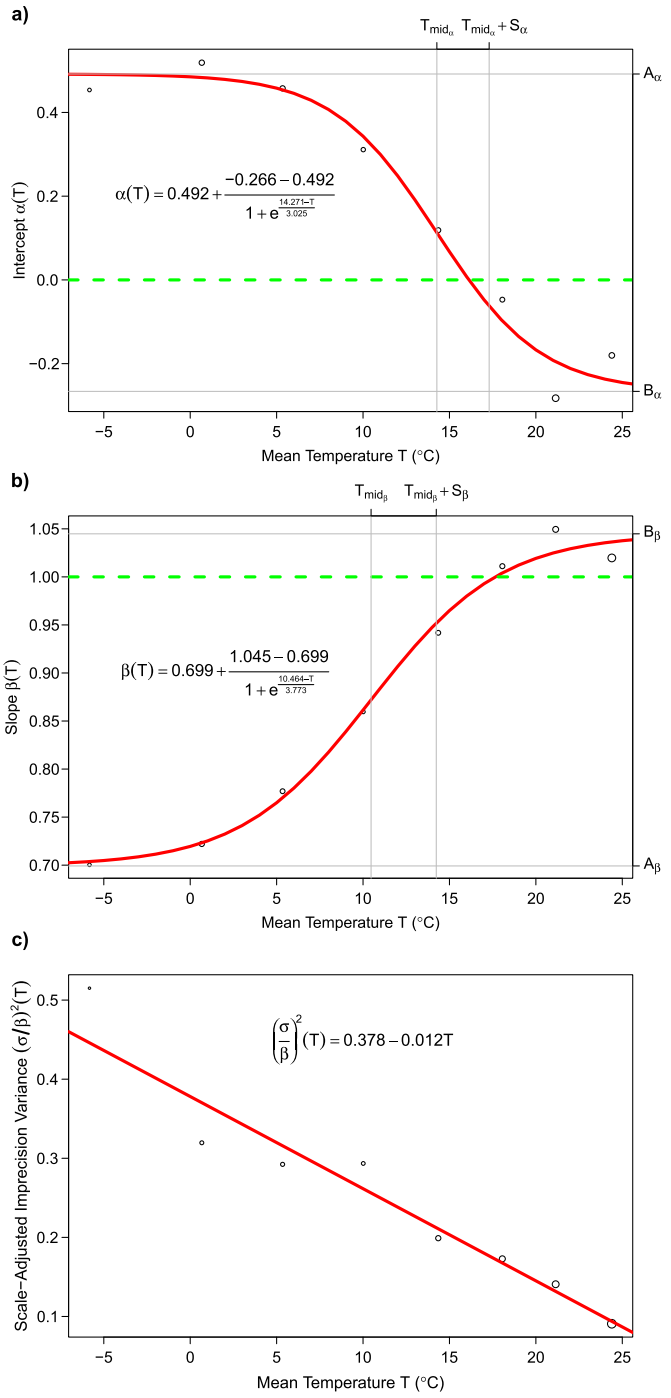


Fig. 5. Model fits for the TEOM/FRM calibration as a function of temperature, T (FRM used arbitrarily as the reference): a) intercepts $\alpha(T)$, b) slopes $\beta(T)$, and c) scale-adjusted imprecision variances $(\sigma/\beta)^2(T)$. The calibration curve is then given by: $\sqrt{TEOM} = \alpha(T) + \beta(T)\sqrt{FRM}$ and the scale-adjusted imprecision standard deviation $(\sigma/\beta)(T) = \sqrt{0.378 + 0.012t}$. Alternatively, based on these functions: $FRM = -\alpha(T)/\beta(T) + 1/\beta(T)\sqrt{TEOM}$ and the imprecision standard deviation is $\sigma_{FRM}(T) = 0.103$.

$$TEOM = \left(\alpha(T) + \beta(T)\sqrt{FRM}\right)^2 \tag{22}$$

or alternatively as:

$$\sqrt{FRM} = -\frac{\alpha(T)}{\beta(T)} + \frac{1}{\beta(T)}\sqrt{TEOM} + \left[\varepsilon_{FRM}(T) - \frac{\varepsilon_{TEOM}(T)}{\beta(T)}\right] \tag{23}$$

with expectation:

$$FRM = \left(-\frac{\alpha(T)}{\beta(T)} + \frac{1}{\beta(T)}\sqrt{TEOM}\right)^2 \tag{24}$$

where

$$\alpha(T) = A_\alpha + \frac{B_\alpha - A_\alpha}{1 + e^{\frac{T_{mid_\alpha} - T}{S_\alpha}}} = 0.492 + \frac{-0.266 - 0.492}{1 + e^{\frac{14.271 - T}{3.025}}}, \tag{25}$$

$$\beta(T) = A_\beta + \frac{B_\beta - A_\beta}{1 + e^{\frac{T_{mid_\beta} - T}{S_\beta}}} = 0.699 + \frac{1.045 - 0.699}{1 + e^{\frac{10.464 - T}{3.773}}}, \tag{26}$$

$$\varepsilon_{TEOM}(T) \sim N\left(0, (a + bT)^2\right) = N\left(0, (0.378 - 0.012T)^2\right) \tag{27}$$

and

$$\varepsilon_{FRM}(T) \sim N\left(0, 0.103^2\right). \tag{28}$$

The calibration curves relating TEOM and FRM as a function of temperature are shown in Fig. 4. As temperature increases, the calibration curve approaches the diagonal line (no bias model).

4.3. Calibration predictions and bias for FRM based on TEOM

The calibration equations previously developed for FRM based on TEOM measurements (and vice versa) (Equations (21)–(26)) are a nonlinear function of the observed TEOM measurement and temperature.

To obtain a clearer understanding of how the bias changes, Table 6 shows the predicted FRM for observed TEOM measurements from 0 to 50 μg/m^3 in steps of 10 μg/m^3 with temperature ranging from −10 °C to 40 °C in steps of 5 °C. Fig. 6 shows the predicted bias (FRM − TEOM) for the same concentration and temperature ranges. The FRM and TEOM exhibit excellent agreement at certain combinations of $PM_{2.5}$ concentration and ambient temperature, as indicated by solid thick zero bias contour lines.

Table 5
Maximum likelihood parameter estimates for modeling the $\sqrt{TEOM}$ calibration relative to $\sqrt{FRM}$ as a function of temperature (T).

Meaning	Symbol	Estimate	Std. Error	t Value	Pr (>|t|)
Intercepts α_T					
Asymptote $T \to -\infty$	A_α	0.492	0.054	9.17	0.0008
Asymptote $T \to \infty$	B_α	−0.266	0.089	−3.00	0.0401
Location (°C)	T_{mid_α}	14.271	1.297	11.00	0.0004
Scale (°C)	S_α	3.025	1.244	2.43	0.0719
Slopes β_T					
Asymptote $T \to -\infty$	A_β	0.699	0.022	32.08	<0.0001
Asymptote $T \to \infty$	B_β	1.045	0.018	58.03	<0.0001
Location (°C)	T_{mid_β}	10.464	0.934	11.20	0.0004
Scale (°C)	S_β	3.773	0.888	4.25	0.0132
Scale-adjusted imprecision variances σ_T/β_T					
Intercept	a	0.378	0.020	18.55	<0.0001
Slope	b	−0.012	0.001	−9.71	<0.0001

Table 6
Predicted FRM measurement based on observed TEOM measurement as a function of temperature.

TEOM value	Temperature (°C)										
	−10	−5	0	5	10	15	20	25	30	35	40
0	0.49	0.48	0.46	0.36	0.16	0.00	0.03	0.06	0.06	0.06	0.06
10	14.53	14.37	13.85	12.50	10.71	10.28	10.67	10.78	10.78	10.77	10.76
20	32.29	31.93	30.72	27.54	22.97	20.83	20.71	20.66	20.60	20.57	20.56
30	50.65	50.08	48.16	43.06	35.52	31.42	30.65	30.40	30.27	30.23	30.21
40	69.33	68.54	65.89	58.83	48.21	42.04	40.55	40.07	39.87	39.80	39.78
50	88.21	87.21	83.82	74.76	61.00	52.67	50.41	49.69	49.42	49.33	49.31

However, it is clear that for observed TEOM values from about 6 to 34 μg/m^3, the TEOM is always biased low relative to the FRM regardless of temperature. This bias is substantial at low temperatures (e.g., 20 μg/m^3 for a TEOM value of 30 μg/m^3 and an ambient temperature of −10 °C) but decreases in magnitude as temperature increases and ambient $PM_{2.5}$ concentration decreases. For TEOM values above 34 μg/m^3, the bias follows a similar trend with temperature, except that it eventually reverses sign (i.e., such that TEOM values are biased slightly high relative to FRM values at sufficiently high temperatures). For TEOM values below 6 μg/m^3, the bias tends to be small regardless of temperature, and it vanishes for certain combinations of TEOM and temperature in a nonlinear fashion. In all cases, the bias between the TEOM and FRM tends to be small (i.e., <1 μg/m^3) for ambient temperatures above 17.65 °C, which is denoted by the vertical dashed line in Fig. 6.

4.4. Comparison of the SEM calibration with ordinary regression

As the correlation between the samplers drops from one and approaches zero, the regression lines increasingly diverge. The calibration curve always falls between the two regression lines and is closer to the regression line that has the independent variable with the lowest imprecision standard deviation. For example, because the FRM has a much smaller imprecision than the TEOM at low temperatures, the regression line for TEOM as a function of FRM is very close to the actual calibration line at these temperatures, whereas the regression line for FRM as a function of TEOM is farther from the actual calibration line. This effect diminishes as the TEOM imprecision standard deviation decreases (i.e., with increasing temperature) and approaches the constant imprecision standard deviation for the FRM. If the two samplers had equal imprecision, the calibration line would be equally distant from both regression lines.

For illustrative purposes, the TEOM and FRM collocated measurements at −5.8 °C were used to compute the ordinary regression curves (on the square root scale and transformed back to the original scale) for FRM as a function of TEOM and vice versa. The deficiencies in doing this have been previously described. As shown in Fig. 7, the regression of TEOM on FRM produces a curve that is not too far from the actual calibration curve, because the FRM (used as the independent variable) has substantially smaller imprecision than the TEOM at this temperature. The typical approach used by the EPA and others (Eberly et al., 2002; Bortnick et al., 2002; Vega et al., 2003; Kashuba and Scheff, 2008), however, is to use the regression of the FRM on the TEOM. This regression curve is substantially different from the actual calibration curve (and the other regression curve). For example, a TEOM measurement of 15 μg/m^3 corresponds to an FRM measurement of about 24 μg/m^3 at −5.8 °C as shown by the calibration curve, giving a relative bias of about 9 μg/m^3. The regression curve for FRM as a function of TEOM predicts an FRM mass concentration of only about 17.5 μg/m^3, resulting

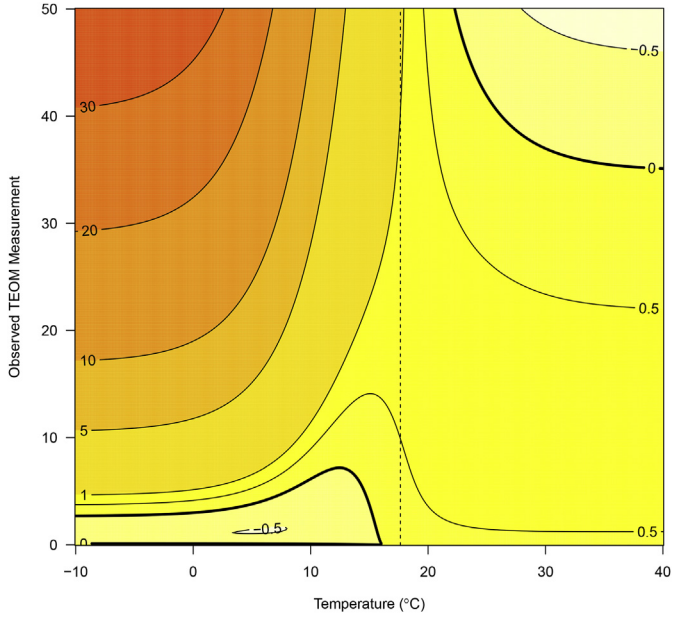


Fig. 6. Bias between observed TEOM measurement and equivalent FRM measurement as a function of temperature. The vertical dashed line at 17.65 °C indicates the temperature at which the bias is a constant bias.

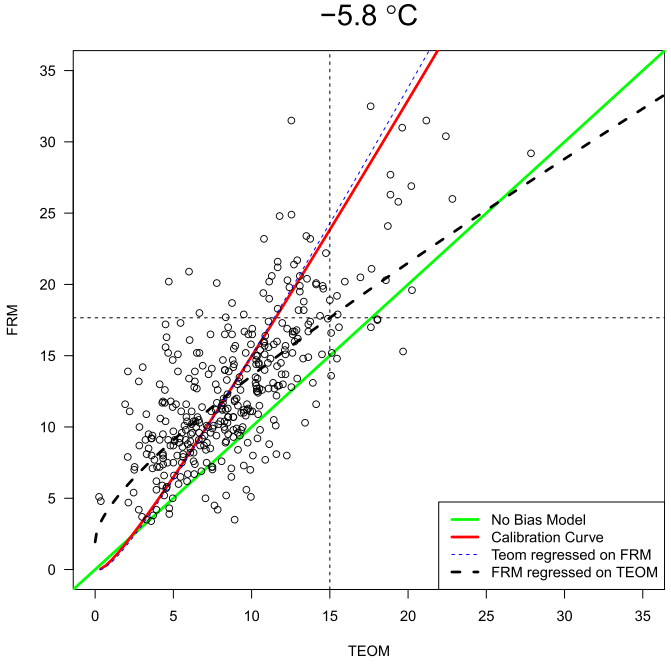


Fig. 7. Comparison of the SEM-derived calibration equation with naive regression equations for TEOM versus FRM and FRM versus TEOM at −5.8 °C.

in an underestimation in the bias of about 6.5 $\mu g/m^3$. In general, this regression curve is a very poor estimate of the actual calibration curve. The fact that the TEOM as a function of the FRM was a better choice in this situation unfortunately cannot be generalized. When the imprecision standard deviations of both samplers are equal, both regression lines will be poor estimates of the calibration curve if there is sufficient random error in the samplers.

5. Conclusions

Although ordinary linear regression analysis commonly has been applied for calibrating $PM_{2.5}$ measurement devices, this approach generally is inappropriate when both measurement devices are subject to random error. Ordinary regression will produce a nearly correct calibration curve when one of the measurement devices is much more precise than the other device, provided that the more precise device is specified as the independent variable in the regression. Using an imprecise measurement device as the independent variable can produce a regression line that is far from the correct calibration curve. Moreover, while ordinary regression can produce an approximately correct calibration line when one device is much more precise than the other, it requires a priori knowledge of the difference in imprecision standard deviations (adjusted for any scale bias) between the two devices. In practice, this is difficult to apply, because if the scale bias and imprecision were already known, the calibration equation would also in effect already be known. Errors-in-variables methods such as Deming regression and RMA regression require similar upfront knowledge of the imprecision standard deviations and are similarly impractical to apply. In addition, all of these regression methods under the best circumstances only use information from two measurement devices and therefore fail to incorporate all the available information in calibrations involving more than two collocated devices.

Structural equation modeling as described in this paper provides a general and flexible methodology to appropriately estimate the calibration curves relating $PM_{2.5}$ mass concentration samplers. This approach uses all available information to simultaneously estimate bias and imprecision and does not require making uninformed assumptions or obtaining external information about the random errors associated with those samplers. In addition, it is possible to include other factors that influence the calibration even if the effect of those factors is non-linear. Finally, the SEM software makes it possible to easily compute calibration equations for most practical applications, to place confidence interval bounds on the calibration curve, and to produce prediction intervals for converting measurements from one device scale to another. It should be emphasized, however, that datasets appropriate for constructing calibration curves must either include collocated measurements from more than two devices or include independent replicates of at least one of the devices when there are only two devices. This assumes that the random errors in each device are statistically independent and uncorrelated. In some circumstances, for example, when one measurement is in part derived from another measurement, the errors will likely be correlated and this will require a larger number of devices and/or independent replicates.

The results presented here demonstrate the successful application of SEM to characterize the relative bias and imprecision among seven collocated $PM_{2.5}$ samplers at an urban monitoring site in Pittsburgh, Pennsylvania. The modeling shows that: (1) the three FRM samplers were considerably more precise than the three speciation samplers and the TEOM sampler, (2) the TEOM sampler showed a significant negative bias relative to the FRM samplers, particularly at low ambient temperature, and the magnitude of this bias decreased non-linearly with increasing temperature, and (3) the speciation samplers showed only small or statistically insignificant bias relative to the FRM samplers, and this bias was not temperature-dependent. This analysis is expanded in a future paper to calibrate data from 97 $PM_{2.5}$ measurement devices at 47 monitoring sites so that these data can be used to provide an unbiased representation of spatiotemporal variations in $PM_{2.5}$ concentration for epidemiologic modeling.

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